

Lab 2 - Technical Communication

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1. The SNOW-V cipher is used for encrypting voice communications in 4G/5G networks. In Algorithm 1, we define the secret key and the initialization vector loading phase of the SNOW-V stream cipher.

Algorithm 1: The key and IV setup phase

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1 procedure KEYIV_SETUP( $K, IV, is\_aad\_mode$ ):
2    $(a_{15}, a_{14}, \dots, a_8) \leftarrow (k_7, k_6, \dots, k_0)$   $\triangleright K = (k_7, k_6, \dots, k_0)$ 
3    $(a_7, a_6, \dots, a_0) \leftarrow (iv_7, iv_6, \dots, iv_0)$   $\triangleright IV = (iv_7, iv_6, \dots, iv_0)$ 
4    $(b_{15}, b_{14}, \dots, b_8) \leftarrow (k_{15}, k_{14}, \dots, k_8)$ 
5    $(b_7, b_6, \dots, b_0) \leftarrow (0, 0, \dots, 0)$ 
6    $R_1, R_2, R_3 \leftarrow 0, 0, 0$ 
7   for  $t = 1 \dots 16$  do
8      $T_1 \leftarrow (b_{15}, b_{14}, \dots, b_8)$ 
9      $z \leftarrow (R_1 \boxplus_{32} T_1) \oplus R_2$ 
10    FSM_UPDATE()
11    LFSR_UPDATE()
12     $(a_{15}, a_{14}, \dots, a_8) \leftarrow (a_{15}, a_{14}, \dots, a_8) \oplus z$ 
13    if  $t = 15$  then
14       $R_1 \leftarrow R_1 \oplus (k_7, k_6, \dots, k_0)$ 
15    end if
16    if  $t = 16$  then
17       $R_1 \leftarrow R_1 \oplus (k_{15}, k_{14}, \dots, k_8)$ 
18    end if
19  end for
20 end procedure
```
